Waste Management 32 (2012) 1116–1121

Contents lists available at SciVerse ScienceDirect

Waste Management

journal homepage: www.elsevier.com/locate/wasman

Effect of organic compositions of aerobically pretreated municipal solid waste on non-methane organic compound emissions during anaerobic degradation

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ARTICLE INFO

Article history:
Received 15 November 2011
Accepted 3 January 2012
Available online 14 February 2012

Keywords:
Non-methane organic compounds
Organic composition
Aerobically pretreated municipal solid waste
Anaerobic degradation

ABSTRACT

Odor pollution caused by municipal solid waste (MSW) treatment plants has become a growing public concern. Although aerobic pretreatment of MSW has advantages in accelerating landfill stabilization, the property of non-methane organic compound (NMOC) emissions from aerobically pretreated MSW (APMSW) during landfilling is unknown. To investigate NMOC emissions from anaerobic degradation of APMSW and to study the impact of organic compositions of APMSW and their decomposition stages, five simulative anaerobic bioreactors (R1–R5) were filled up with APMSW of different original organic compositions in a laboratory. For NMOC analysis, samples were collected from the gas that accumulated separately during two successive independent stages of the whole experiment. The results showed that the cumulative quantities of NMOCs from R1 to R5 were 1.11, 0.30, 0.18, 0.28, and 0.31 mg/kg DM, respectively, when volatile solid was degraded by 34.8–47.2%. As the organic content of the original waste was lower, the proportion of NMOCs generated in the early stage of anaerobic degradation became higher. Multiple linear regression analyses of the relationship between the quantities of degraded organics and generated NMOCs showed that lipid and protein have a strong effect on NMOC amount. The effect of lipid on NMOC quantity lasts longer than that of protein. This observation suggests that controlling the lipid and protein contents in MSW can reduce the odor from landfills.

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1. Introduction

Conventional landfilling is a common method adopted worldwide to dispose of municipal solid waste (MSW). During the anaerobic degradation of MSW, most organic matter is broken down by microorganisms into simple compounds, leading to the formation of landfill gas. CO_2 and CH_4 are the dominant components in landfill gas. Trace species, such as mercury, arsenic, and some organic compounds, are often found abundantly in landfill gas relative to other environmental sources (Kim and Kim, 2002; Feng et al., 2004; Huang et al., 2009). Although non-methane organic compounds (NMOCs) comprise less than 1% (V/V) of landfill gas, they may create a disproportionate environmental burden and potential risk to human health (Allen et al., 1997). For example, emissions of carcinogens, such as benzene and vinyl chloride, pose a potential threat to workers and local inhabitants (El-Fadel et al., 1997; Durmusoglu et al., 2010). Furthermore, chloro- and fluorohydrocarbons emitted from landfills seriously corrode combustion engines that utilize landfill gas (Dernbach, 1985), deplete the ozone layer, and cause climate change (Molina and Rowland, 1974; Wallington et al., 1994).

The reported yields of NMOCs varied widely according to the nature of waste and the experiment condition (Thomas and Barlaz, 1999; Smet et al., 1999; Staley et al., 2006; Wang and Wu, 2008). The yield of total terpenes from orange wastes was up to 11,000 mg/kg DM (Wang and Wu, 2008), and the NMOC yields of refuse, paper, yard waste, and food waste were investigated to be 151, 16, 38, and 221 mg C/kg DM, respectively (Staley et al., 2006). It was also found that NMOC release during aerobic decomposition of refuse and yard waste was larger than that in anaerobic systems (Staley et al., 2006). However, Komilis et al. (2004) detected a VOC yield of food waste as low as 0.33 mg/kg DM under controlled aerobic conditions, approximately four orders of magnitude less than values reported by Staley et al. (2006).

Knox (1990) compared NMOCs released from three main phases of decomposition, and found that total NMOC concentration initially increases and then steadily decreases. Moreover, the author also investigated some categories of NMOC concentrations among waste degradation processes. Alcohols and halogenated hydrocarbons are the main components of NMOCs emitted from young refuse (Knox, 1990). Deipser and Stegmann (1994) found that the highest concentrations of volatile chlorinated and fluorinated hydrocarbons occur during the acid phase in a landfill. Kim (2006) similarly concluded that the proportion of reduced sulfur compounds is larger in young landfills than in old landfills.

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0956-053X/$ - see front matter © 2012 Elsevier Ltd. All rights reserved.
doi:10.1016/j.wasman.2012.01.005

Aromatics and alkanes are the primary components emitted from middle-aged and old refuses (Knox, 1990).

Sources of NMOCs in landfill gas include the anaerobic biodegradation intermediates and the volatilization of hazardous organic compounds present in MSW (Thomas and Barlaz, 1999). The decomposition by-products of readily degradable organic matter are oxygenated compounds, alkanes, and alkenes (Smet et al., 1999). Aromatics may also be present in food waste because most aromatics are lipophilic compounds that can be absorbed by lipids (He et al., 2010). Yard waste, fruit peels, and vegetable leavings are particularly high in terpenes (Wang and Wu, 2008). According to Allen et al. (1997), domestic waste releases more alkanes, alcohols, and ketones than trade waste. Consumer products from the chemical industry contain chlorinated compounds that are directly volatized in landfills (UK Environment Agency, 2004). Young and Parker (1983) found that the gas evolved by industrial landfills is considerably richer in hydrocarbons and solvents, including halocarbons, than that from landfills receiving only domestic waste.

Aerobic pretreatment of MSW can not only reduce mass of waste but improve landfill behavior. When aerobically pretreated MSW (APMSW) is buried in a landfill, the leachate pollution load and the biogas production potential will be diminished (Kuruparan et al., 2003; Lornage et al., 2007; Scaglia et al., 2010; Tambone et al., 2011), the stabilization process will be enhanced (Leikam and Stegmann, 1999; Zach et al., 2000), and the intensity of the NMOC emissions will be reduced (Zhang et al., in press). To our best knowledge, however, little information is available on the effect of organic compositions of APMSW on NMOC characteristics at different degradation stages. This information is important because it is helpful for determining the way and the extent of pretreatment from perspective of odor control. The objectives of this study are as follows: (1) to investigate the NMOC emissions from APMSW with various original organic contents by measuring NMOC yields and components at two different stages of the anaerobic process and by analyzing emission intensities, and (2) to correlate the degraded organic contents with the NMOC yields.

2. Materials and methods

2.1. Materials

The original waste A1, approximately 500 kg of MSW was obtained from a landfill in Beijing, China. The waste was aerobically treated to various extents, obtaining five waste samples of different organic compositions (Table 1). The detailed aerobic pretreatment process for the five wastes is introduced in literature (Zhang et al., in press).

2.2. Experimental setup

The reactors, made up of high-density polyethylene, each had an internal diameter of 500 mm and a height of 750 mm (Fig. 1). The operation modes are shown in Table 2. The reactors R1–R5 were filled with A1–A5, respectively.

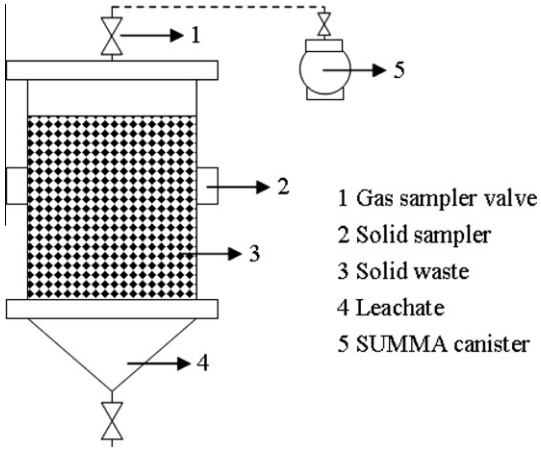


Fig. 1. Scheme of the reactor.

Table 2
Operation modes of the five reactors.

Reactors	Materials	Weight (kg)	Height (cm)	Time (d)	
				Stage I	Stage II
R1	A1	77.5	53.5	1st–127th	128th–359th
R2	A2	63.2	41.0	1st–97th	98th–329th
R3	A3	61.2	37.5	1st–77th	78th–309th
R4	A4	58.2	36.5	1st–77th	78th–309th
R5	A5	57.0	33.5	1st–77th	78th–309th

2.3. Sampling and analytical methods

The gas accumulated in the reactors was collected separately at the end of stage I or II using a previously evacuated SUMMA canister and then analyzed for the determination of its NMOC composition. Solid samples were obtained immediately after gas sampling from the two side samplers and the top of the reactors.

Moisture content (wet basis) was determined by heating the ground sample at 105 °C for 24 h. Volatile solid (VS) was determined by ashing the dry samples at 550 °C to a constant weight in a muffle furnace (SX2-8-10; NuoJi Instrument Co. Ltd., Changzhou, China). Before the organic components were analyzed, the waste sample was first dried at 105 °C, and then shredded using a shredding machine (LK-500; Nongfeng Instrument Co. Ltd., Zhongshan, China). Protein was determined as Kjeldahl *N* times a factor of 6.25 (APHA, 1998). Lipid was determined through extraction using ether (AOAC, 1990, method no. 920.39). The remaining organics were called "others," which were calculated by subtracting the sum of lipid and protein from VS.

Total carbon (TC) and total nitrogen (TN) were measured using a carbon–hydrogen–nitrogen analyzer (Equipment CE 440; EAI USA).

Table 1
Characteristics of the five kinds of waste for anaerobic degradation.[a]

Waste	Moisture (%)	VS (%)	TC (%)	TN (%)	Lipid (%)	Protein (%)	Others[b] (%)
A1	61.8	65.1	34.9	2.1	7.8	13.1	44.2
A2	70.2	49.6	22.6	1.8	0.6	11.1	37.9
A3	70.0	45.2	20.5	1.5	0.3	10.2	34.7
A4	71.0	38.2	18.4	1.2	0.3	9.8	28.1
A5	70.0	30.2	15.1	1.1	0.1	9.4	20.7

[a] A1 indicates the original waste obtained from a landfill site; A2, A3, A4, and A5 represent the waste produced from aerobic pretreatment of A1.
[b] Others are determined by subtracting lipid and protein from VS.

A 6 L SUMMA canister was used to collect the gas sample. A 150 ml gaseous sample was extracted from canister (sampling flow was set at 50 ml/min) and adsorbed by adsorption tube (Tenax/Carbograph 1TD/Carboxen 1000). The sample was concentrated into a cold trap containing two beds of sorbent, i.e., Tenax TA and Carbograph, supported by quartz wool before it was analyzed by gas chromatography coupled with flame ionization detection (FID). The measurement setup consisted of a flow controller, a thermal desorber (Markes International, UK), and a gas chromatograph (GC6890; Agilent, USA) equipped with FID and jet-modulated parts (KT2001; Zeox, USA) for comprehensive two-dimensional gas chromatography (GC × GC) in accordance with the method used by Xu et al. (2003). The sampling and desorption control software (Markes International, UK) and ChemStation (Agilent, USA) were installed on a personal computer and used to control the sampling/thermal desorption system and the gas chromatograph, respectively.

The thermal desorption temperature was 300 °C. Both cold and hot jets of nitrogen gas were used. The jet-modulated system was operated for 6 s at a cryotrap temperature of −20 °C.

The column set used for GC × GC equipment was composed of an HP-5MS (5% phenyl and 95% dimethylpolysiloxane phase) primary column (30 m × 0.25 mm I.D. × 0.25 μm film thickness) directly coupled to a DB-WAX (polyethylene glycol phase) secondary column (1.05 m × 0.1 mm I.D. × 0.1 μm). The temperature program for the primary column began at 35 °C, held for 8 min, and then rose to 221 °C at 3 °C/min. For the secondary column, the temperature was initially held at 30 °C for 8 min, and then raised to 210 °C at 3.3 °C/min and held for 7.5 min. The transfer line column for the system was a 1.5 m deactivated fused silica column with a 0.1 mm I.D.

USEPA standard gases (EPA TO-15) were used for identification of NMOCs. TO-15 (1000×10^{-9} V/V) was diluted with high purity N_2 to the concentration 20×10^{-9} V/V. The calibration curve using external standard method was prepared with the diluted standard gases of six different samplers' volumes (30, 60, 120, 240, 480, and 720 ml). The r^2 of the calibration curve exceeded 0.99. The mean standard deviations of the first and second dimension retention times of the target compounds were less than 0.6 s and 0.02 s, respectively. The method detection limits for alkanes, aromatics, oxygenated, and halohydrocarbon were 4.23, 5.17, 3.41, and 7.80 ng, respectively.

2.4. Statistical analysis

Statistical analysis was performed using SPSS 18.0 for windows (SPSS Inc., 2009). Multiple linear regression analysis was performed by enter. In the course of enter regression analysis, the probability values of F were set as $\leqslant 0.05$ and $\geqslant 0.1$ for entering and removing a variable, respectively. Correlation and regression analyses were included to examine NMOCs and the three kinds of organics, i.e., lipid, protein, and others, measured in the whole experimental period, i.e., stages I and II, respectively.

3. Results and discussion

3.1. Organic degradation

Table 3 presents the fractional degradation of organic matter for the reactors R1–R5 in stages I and II. In stage I, lipid in R1 and R2, protein in R5, and others in R3 and R4 degraded to the highest extent. Others had the highest fractional degradation in stage II for all reactors except R2, varying from 12.4% to 35.4%. During the whole experimental period, the leachate pH varied within the range of 5.1–5.6, suggesting that A1 was still in the hydrolysis acidification stage. Considerable acidification inhibits the further conversion of organics. For the reactors R2–R5, the leachate pH was above 6.7 and no serious acidification occurred during both stages. The total amounts of VS degradation for R1, R2, R3, R4, and R5 were 34.8%, 47.2%, 41.1%, 39.8%, and 42.8%, respectively.

Table 3
Fractional degradation of organics in stages I and II (%).[a]

Reactors	VS		Lipid		Protein		Others	
	I	II	I	II	I	II	I	II
R1	18.4	16.4	78.2	10.3	24.0	5.4	6.2	20.7
R2	33.5	13.7	51.5	8.3	14.9	18.8	38.6	12.4
R3	16.1	25.0	5.1	24.4	8.2	4.5	18.5	31.1
R4	21.7	18.0	1.9	19.4	10.9	10.5	25.7	20.6
R5	15.4	27.4	5.0	20.2	21.9	9.8	12.5	35.4

[a] Reported data represent percentages of degraded organics relative to the original ones in terms of VS, lipid, protein, and others.

Moreover, the compositions of the degraded organics were investigated (Fig. 2). Others dominated the degraded organics for all reactors in both stages except for R1, where lipid was the dominant degraded component in stage I. The normalized amount of degraded protein always overweighed that of lipid in both stages for reactors R2–R5.

3.2. Yields of NMOCs

The cumulative quantity of NMOCs during stages I and II for each reactor are presented in Fig. 3. The cumulative weights of the NMOCs generated in stages I and II were 0.48 and 0.63, 0.13 and 0.17, 0.07 and 0.12, 0.17 and 0.11, and 0.16 and 0.15 mg/kg DM for R1–R5, respectively. The current results were significantly different from the previously reported values (Thomas and Barlaz, 1999; Staley et al., 2006; Wang and Wu, 2008), probably because of differences in the experimental scale and characteristics of waste. The partial NMOC yields in stage II were always larger than in stage I from R1 to R3, where the whole NMOC yields decreased. In contrast, the total NMOC yields slightly increased from R4 to R5, where the partial NMOC yields in stage I were larger. A possible explanation for this result may be that NMOC yield was affected by the various degraded components and the conversion paths of organics. Staley et al. (2006) investigated the NMOCs released from paper, yard waste, and food waste under anaerobic conditions, and

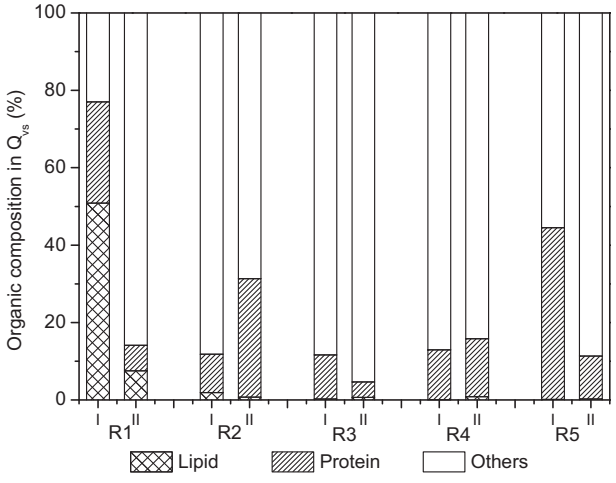


Fig. 2. Degraded organic composition for the five reactors in stages I and II. Q_{vs} is the quantity of degraded volatile solids (VS) in stage I or II. Others are determined by subtracting lipid and protein from VS.

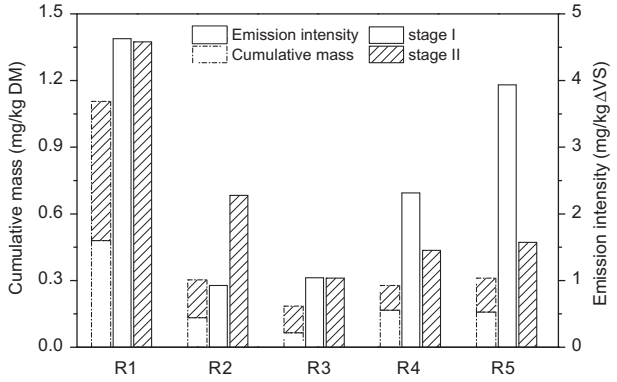


Fig. 3. Yields and emission intensities of NMOCs from the five reactors in stages I and II.

found that NMOC yields from food waste were 5.8 times that of yard waste and 13.8 times that of paper. The waste A1 initially contained considerably higher lipid than the other wastes. The percentage of degraded lipid in the total organics for R1 was higher than that for the other reactors. Moreover, serious acidification in R1 inhibited microbial activities, and the organic intermediates were accumulated.

To obtain a comparable situation among the reactors, the NMOC emissions from R1 to R5 were normalized to emission intensities of NMOCs by dividing the cumulative NMOC emissions by the quantities of organic matter removed during the anaerobic period (Q_{VS}), which are also shown in Fig. 3. The emission intensities of NMOCs in stages I and II from R1 to R5 were 4.0 and 4.0, 0.8 and 2.0, 0.9 and 0.9, 2.0 and 1.3, and 3.4 and 1.4 mg/kg Q_{VS}, with ratios of 1.01, 0.41, 1.00, 1.59, and 2.50 between them, respectively. In general, as the organic content of the original waste became lower, the proportion of NMOCs became higher at the early stage of anaerobic degradation. A comparison of Figs. 2 and 3 shows obvious coarse correlation between NMOC intensities and the degraded amount of lipid or protein or both in stage I but not in stage II. However, a simple collection of all the other organics as a uniform "others" group neglects the possible differences among components such as starch and fibers in releasing NMOCs. Further research is necessary to investigate these details.

3.3. Composition of NMOCs

The NMOCs measured in this study can be grouped into five categories: oxygenated compounds (i.e., ketones, alcohols, aldehydes, acids, and esters), alkanes, alkenes, aromatics, and chlorinated compounds, similar to the method used by Ran et al. (2009). The compositions of NMOCs generated from these wastes during stages I and II are shown in Fig. 4. The dominant NMOC categories were either oxygenated compounds or alkenes. In both stages, the proportion of oxygenated compounds decreased and that of alkenes increased from R1 to R5. A comparison of the compositions of NMOCs in stages I and II showed that, with time, the percentage of oxygenated and chlorinated compounds decreased, whereas the percentage of hydrocarbon compounds such as alkanes and alkenes increased. The results agree with the conclusions of Knox (1990) and Deipser and Stegmann (1994). The trends of dominant NMOC categories from R1 to R5 in stages I and II may be caused by the fractional degradation of organic matter. The oxygenated compounds were derived from the decomposition by-products of readily degradable organic matter. Based on the degradation efficiencies in Table 3, the highest degradation for lipid, protein, and others in stage I occurred in R1 and R2. For R2–R5 the main degraded organic composition was others, as shown in Fig. 2. In stage

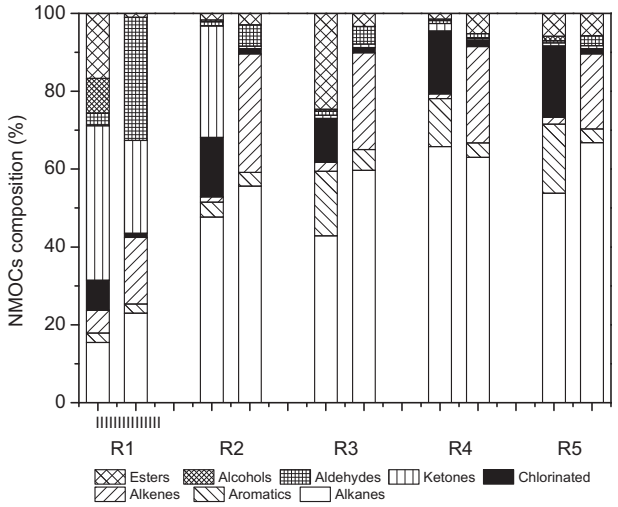


Fig. 4. Composition of released NMOCs from the five reactors in stages I and II. Esters, alcohols, aldehydes, and ketones are classified as oxygenated compounds; alkenes, aromatics, and alkanes are classified as hydrocarbon compounds.

II, the domain compositions of the degraded organics for all of the reactors were others, whereas the fraction of oxygenated compounds in NMOCs was very low (less than 10%), except for R1 (56.5%). A possible explanation for the occurrence of dominant NMOC categories may be the presence of different types of organic components, such as starch and fibers, in the waste. The structures of these components are different in terms of the shape of the bond that connects their glucose molecules to one another, resulting in different degradation rates. Starch, as a kind of readily degradable organic matter, is degraded earlier and easier than fiber. Therefore, the main degraded components of others were fiber for R2–R5 and starch for R1 in stage II.

Nagata (2003) investigated the odor threshold distribution for NMOCs. The author found that the odor threshold is 10–100 ppb for oxygenated compounds, 100 ppb to 1 ppm for hydrocarbons, and 1–10 ppm for chlorinated compounds. Humans are normally more sensitive to NMOCs with lower odor thresholds. With a decrease in organic contents of initial wastes and with the degradations, the NMOC yields decreased and the percentage of oxygenated compounds in NMOCs decreased. Thus, the odor may be weakened, to the point that it is less perceptible for humans.

The main components of the total 89 NMOC species in stages I and II were also investigated. Fig. 5 shows the individual components whose mass contents are more than 5% of total NMOCs during stage I or II. As organic contents of initial wastes decreased, the mass contents of 2-butanone, ethyl acetate, ethanol, and acetone in stage I and the mass content of phenol in stage II declined, while the mass contents of pentane and 1,2-dichloroethane in stage I and the mass contents of hexadecane in stage II increased.

3.4. Correlation analysis between organics and NMOCs

The correlation analysis showed that lipid had significant correlation with NMOCs in stages I and II. In comparison, protein had significant correlation with NMOCs only in stage I, and neither with others. Table 4 summarizes the correlations between the degraded lipid (X_1) and protein (X_2), and the released NMOCs (Y) by enter regression analysis. Based on the partial F test and the value of R, strong linear relationships were found between NMOCs and lipid and/or protein. The effect of lipid on NMOC quantity probably lasted longer than that of protein. Therefore, controlling the lipid and protein contents of MSW for landfill disposal can be an effective option for odor reduction, especially for some Asian countries

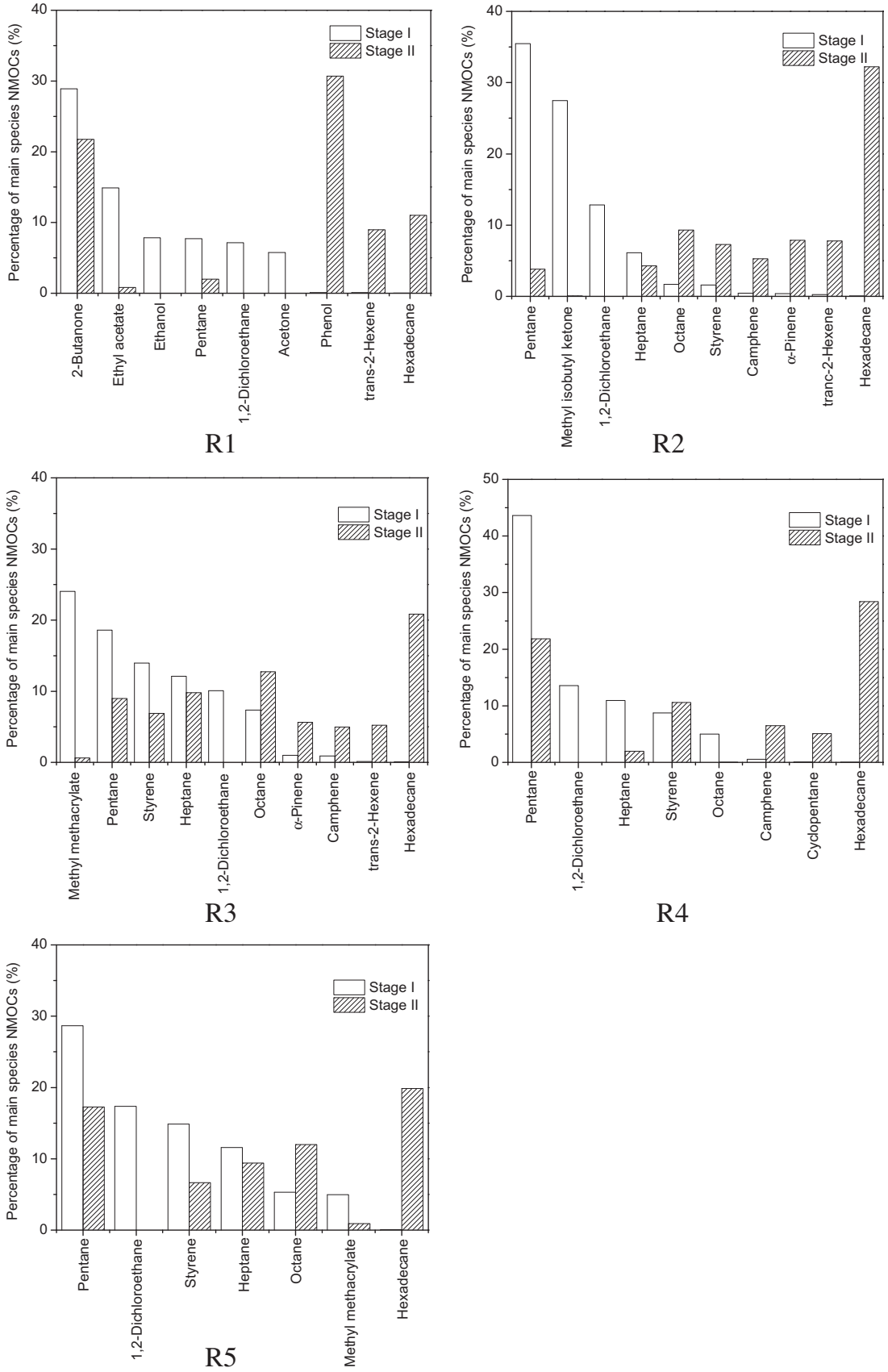


Fig. 5. Dominant components of NMOCs in stages I and II. Components are expressed at the descending order of mass content in stage I.

and areas where the MSW contains relatively high contents of food waste.

However, the effect of others on NMOCs cannot be excluded because what the regression and equations represented was not

Table 4
Multiple linear regression between the quantity of organics and NMOCs.[a]

Stages	Equations	*R*	*p*
I	$Y = 5.1X_1 + 4.0X_2 + 1.2$	0.995	0.011
II	$Y = 46.1X_1 + 1.5$	0.996	0.000

[a] *Y*: quantity of NMOCs, mg; X_1: quantity of degraded lipid, kg; X_2: quantity of degraded protein, kg; *R*: correlation coefficient; *p*: probability against the null in that case.

a real understanding of reaction mechanism but rather an abstractive relationship of figures. Both amounts of NMOC emissions and degraded organics of R1 were considerably larger than those of the other reactors at either stage. This experimental condition played a considerably linearizing role in the regression.

4. Conclusions

The yields and compositions of NMOCs from five kinds of APMSW with various organic contents were investigated in the processes of anaerobic degradation. With total VS degradation of 34.8–47.2%, the emission intensities of NMOCs from R1 to R5 were 8.0, 2.8, 1.8, 3.3, and 4.8 mg/kg Q_{VS}, with ratios of 1.01, 0.41, 1.00, 1.59, and 2.50, respectively, between stages I and II. The dominant categories of NMOCs changed from oxygenated compounds (e.g. 2-butanone, ethyl acetate, ethanol, acetone and phenol) to alkanes (e.g. Hexadecane and pentane) when the readily degradable organic content of waste decreased. Based on the correlation analysis, lipid and protein have significant correlations with NMOC quantity, and the effect of lipid on NMOC emission lasts longer than that of protein. Therefore, controlling the lipid and protein contents of initial MSW will be helpful in reducing odor in landfills.

Acknowledgements

The authors acknowledge the financial supports provided by Beijing Science and Technology Project (No. D09040600380901) and Special Fund of Environmental Protection Research for Public Welfare of China (Nos. 201109035 and 201209022).

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